

# moments

observation map: every coupling, every seed, underlying main Fig. 1

seed 0 1 2 3 4 5

circular moments  $m_1$ ,  $m_3$  of the peer phases ·  $N=17$  oscillators, 100 steps · gpt-5.4-mini · 24 runs, 40,800 calls, 100.0% valid

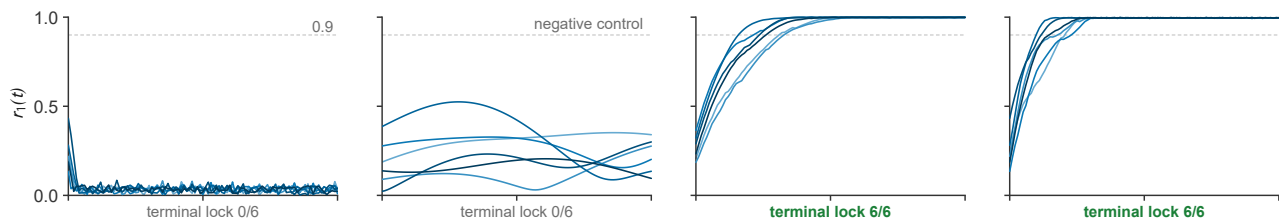
$K = -0.15$

$K = 0$

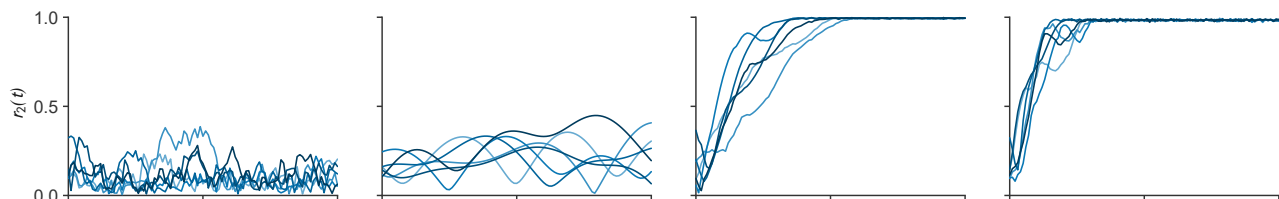
$K = 0.08$

$K = 0.15$

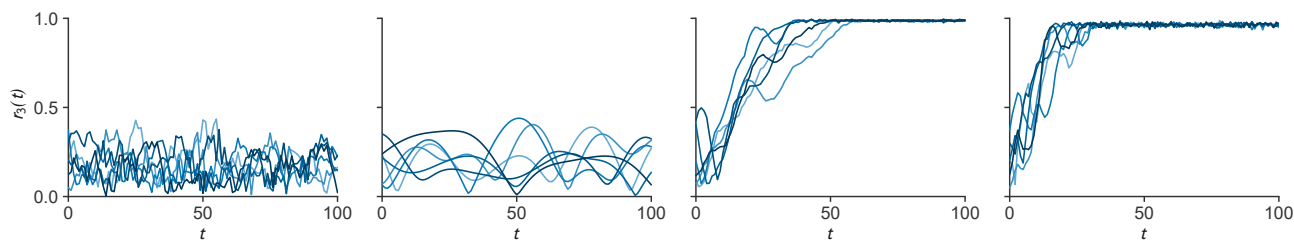
## a Polar order



## b Second harmonic



## c Third harmonic



## d Final phase configuration at $t = 100$ , with final $r_1$ printed under each ring

